TEAM VAST

Anya — Shruti — Trischa

Won 2nd Place National (India) - Score ~ 470-500 points

Anya Agarwal

Anya led **hardware integration** and **navigation development**, including **PID wall-following,** heading correction, ramp detection, and **OpenMV-based victim detection.** She has **experience of over 8 years** in robotics including **multiple languages** and has led multiple teams for **international competitions.**

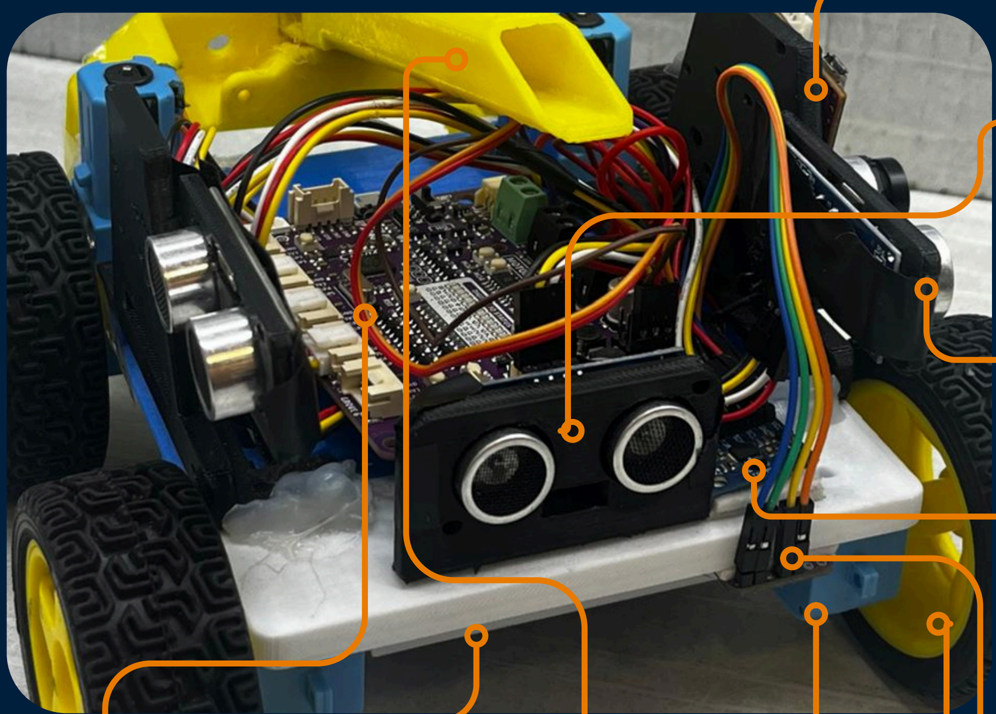
Shruti Purumala

Shruti developed Samarth's **DFS/BFS navigation** algorithms, heading correction, and cell realignment system. She also managed **overall software integration** and has experience in **Python, C++, and computer vision** using OpenCV.

Trischa Taneja

Trischa led Samarth's **mechanical design**, creating the **custom Fusion 360 chassis**, component mounts, sensor cutouts, and the dual-purpose rescue kit deployment mechanism.

HARDWARE



OpenMV H7 + Camera: Detects victims and provides visual information for navigation.

Front Ultrasonic Sensor (1): Measures distance to walls and obstacles ahead.

Side Ultrasonic Sensor (2- left & right): Maintains wall alignment and detects openings.

Gyroscope: Tracks robot orientation and turning angles for accurate movement.

Color Sensor: Detects floor tiles, checkpoints, and special maze zones.

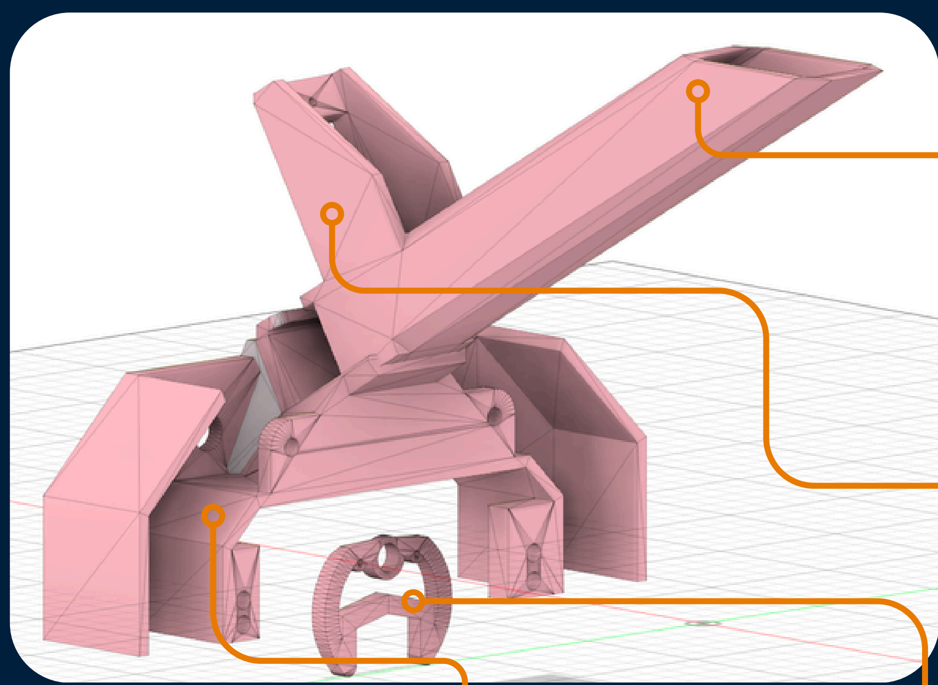
Wheels (4): Provide traction and mobility throughout the maze.

Main Controller (Cytron 2040): Main processing unit that reads sensors, executes algorithms, and controls robot movement.

Battery: Powers all electronic and mechanical systems.

Rescue Kit Dropper Mechanism: Automatically deploys rescue kits when a victim is detected, ensuring accurate and reliable delivery.

Motors (4)(2 right and 2 left): Drives the left and right wheel during movement and turning.

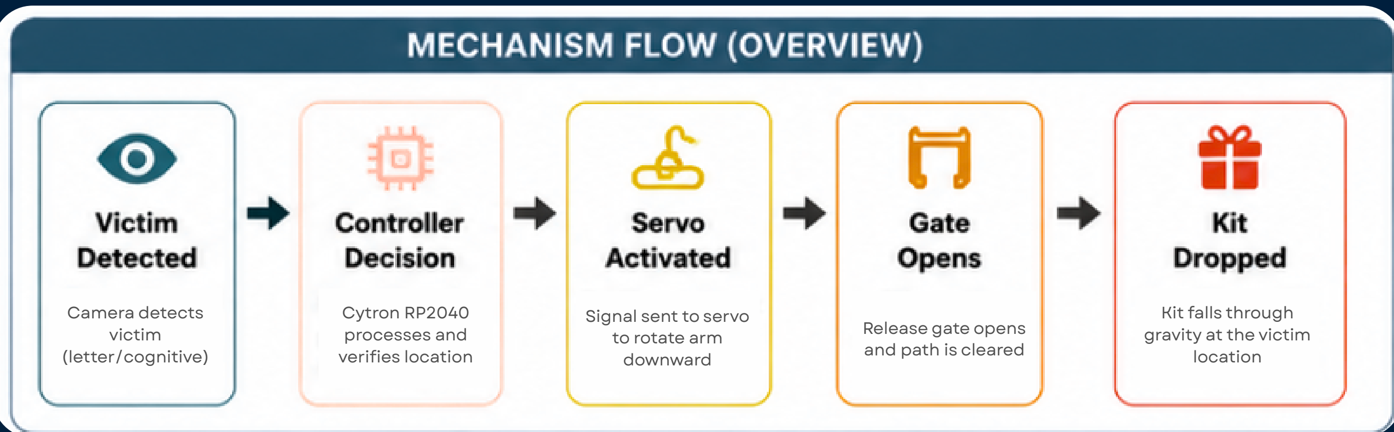


Storage for rescue kits: Stores up to 8 rescue kits and guides them toward the release mechanism using gravity. The servo below will block the rescue kits form dropping into the dropping chute.

Servo: The servo will be mounted here and will be attached with thearm. Rotates the deployment arm to release rescue kits when a victim is detected

Release Arm: Rotates based on victim detection to release a specific number of rescue kits. to position and release rescue kits accurately.

Rescue Kit Drop Chute: Guides rescue kits from the release mechanism to the ground for accurate victim aid delivery. It has 2 ways for ease and so the kit doesn't get stuck.



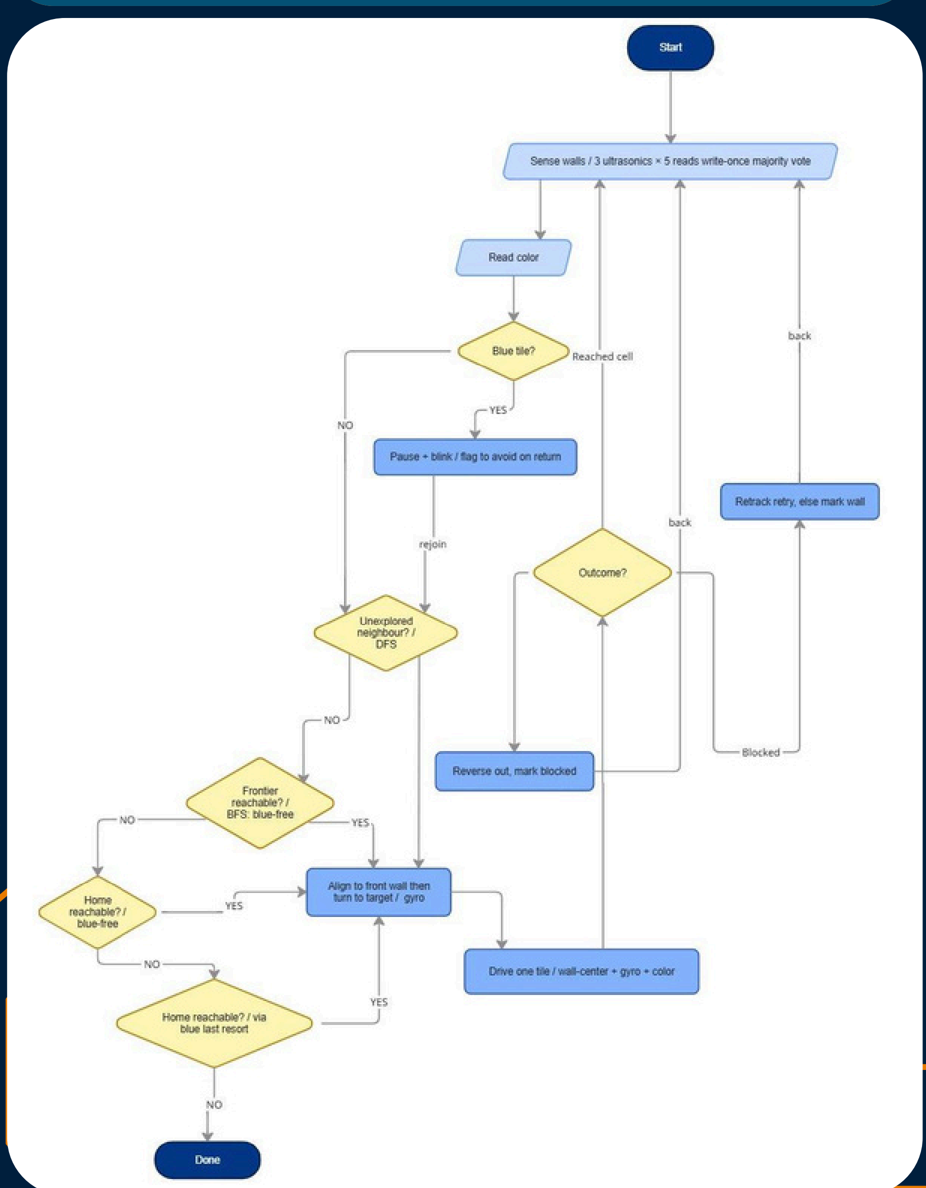
SOFTWARE

Programming Environment:

Language: C++, Python
Controller: Cytron 2040
Libraries Used:

- **MPU6050_LIGHT:** Processes IMU data for orientation and turn tracking.
- **Adafruit_TCS34725:** Reads color sensor data for tile and checkpoint detection.
- **Adafruit_NeoPixel:** Controls RGB LEDs for robot status and victim indication.

Entire Software Overview Diagram- BFS & DFS

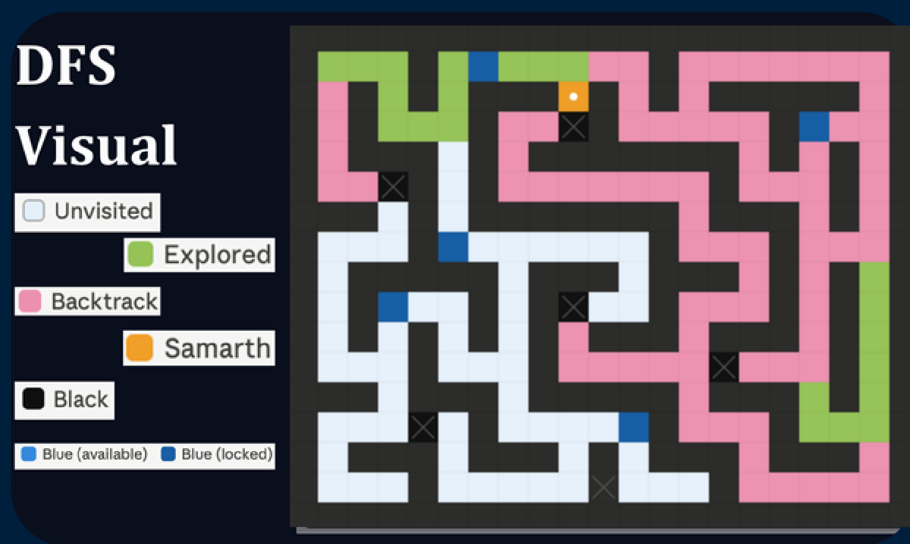


Depth-First Search (DFS) Exploration:

Depth-First Search (DFS) is used to systematically explore the maze. The robot prioritizes unexplored paths, marking each tile as visited and storing junction information. When a dead end is reached, the robot backtracks to the nearest unexplored junction and continues exploring until the entire maze has been mapped.

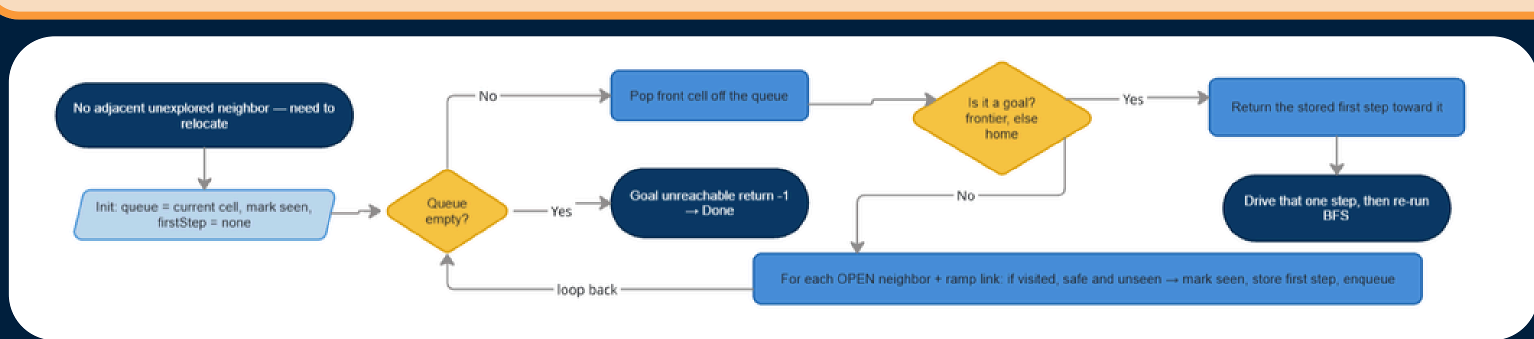
Key Features

- Explores every reachable path.
- Prevents repeated exploration of visited areas.
- Uses backtracking to recover from dead ends.
- Builds a complete map of the maze for later navigation.



Breadth-First Search (BFS) Recovery & Path Planning:

When no unexplored paths remain nearby, Samarth uses Breadth-First Search (BFS) to find the shortest route through previously explored safe cells. The algorithm searches the mapped maze, identifies the nearest unexplored frontier or the start tile, and generates an efficient path to reach it. This allows the robot to recover from dead ends, continue exploration, and reliably return to the starting position after completing the maze.



Key Features

- Allows blue-tile recrossing avoidance
- Recovers efficiently from dead ends.
- Routes the robot to te nearest exploration frontier.
- Guarantees a return path to the start tile.

Victim Detection System

Samarth uses an OpenMV H7+ camera running an Edge Impulse machine learning model to autonomously identify victims. Images captured by the camera are processed onboard in real time, allowing the robot to classify victims, trigger LED indicators, and initiate rescue kit deployment without external computation.

Key Features

- Real-time onboard image processing.
- Edge Impulse machine learning model.
- Autonomous victim classification.
- Triggers rescue kit deployment.
- Provides LED-based victim indication.